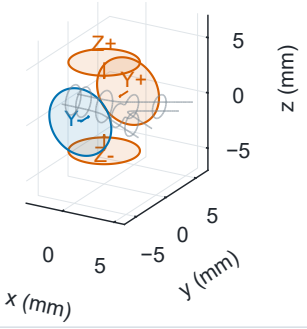
a PZT control setup

B2 | Ideal acoustic sources



$$u_i = s_i A_i e^{i\phi_i}$$

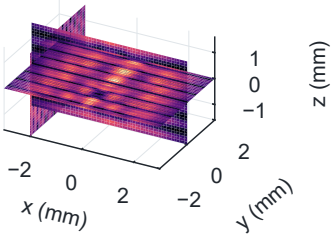
Element	A (kPa)	ϕ (deg)	State
Y+	3.82	180	ON
Y-	15.29	0	ON
Z+	3.82	180	ON
Z-	3.82	180	ON

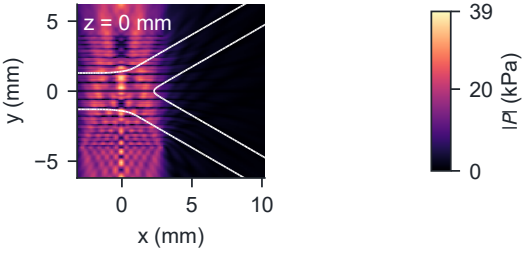
f = 2.5 MHz | Disc diameter = 6 mm

b Three-dimensional acoustic field

Orthogonal slices

k-Wave basis superposition



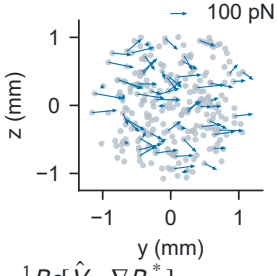


$$P(\mathbf{x}) = \sum_i u_i H_i(\mathbf{x})$$

c Microbubble acoustic forces

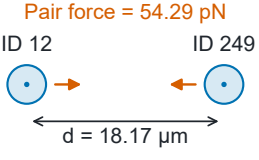
Cycle-averaged; t = 0

Primary Bjerknes force



$$\mathbf{F}_{P,i} = -\tfrac{1}{2} Re[\hat{V}_{i,1} \nabla P_i^*]$$

Secondary Bjerknes force

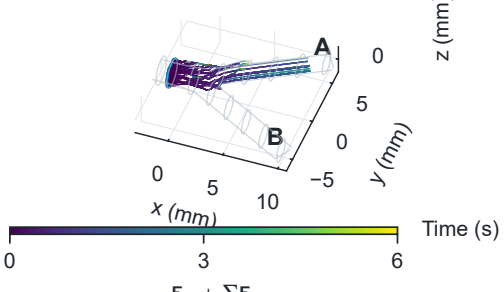


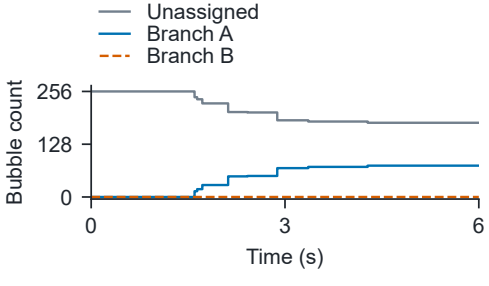
$R_0 = 1.78$ µm; true pair geometry; t = 0

$$\mathbf{F}_{S,ij} = \frac{\rho \langle \dot{V}_i \dot{V}_j \rangle}{4\pi d_{ij}^2} \mathbf{e}_{ij}$$

d Lagrangian particle tracking

256 bubbles; 0-6 s





$$\mathbf{v}_i^{\text{free}} = \mathbf{u}_{\text{flow}} + \frac{\mathbf{F}_{P,i} + \sum_{j \neq i} \mathbf{F}_{S,ij}}{6\pi\mu R_i}$$

RK2 + geometric constraints